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REVIEW

A review of preferential water flow in soil science

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Abstract: A better understanding of preferential water flow is important because water-related crises, i.e., water scarcity and security, are strongly associated with water retention rates in different landscapes. This review aims to evaluate significant advances in the main themes of preferential water flow to establish the inconsistent roles of preferential water flow in eco-hydrology and suggest promising areas for future work. Results showed that preferential water flow studies have made significant advances in our understanding of certain parameters functioning at multiple scales but that most studies focus on preferential water flow in the vadose zone, whereas few studies on the soil surface. Preferential water flow can have a positive effect on averting water crises, such as when it affects surface runoff soil erosion, soil formation as ecosystem services, nutrient cycling in root zone, and overall water regulation of the water cycle. Conversely, preferential water flow can have a negative effect on eco-hydrological issues via slope stability, gully erosion, geological disaster, waste treatment, water supply in root zone, and food production. Our review concludes that more information is required on preferential water flow before we can assess its role in mitigating water-related crisis events.

Key words: preferential water flow, water, soil hydrology, soil.

Résumé : Mieux comprendre l'écoulement préférentiel de l'eau est essentiel, car les crises associées à l'eau (pénurie et sécurité) présentent des liens étroits avec le taux de rétention de l'eau dans différentes formes de relief. Par leur analyse, les auteurs voulaient évaluer les principaux progrès réalisés dans l'étude de l'écoulement préférentiel de l'eau en vue de vérifier les rôles incohérents de ce phénomène en éco-hydrologie et de suggérer les futurs champs de recherche prometteurs. Les résultats montrent que les études sur l'écoulement préférentiel de l'eau ont accompli d'importants progrès qui ont abouti à une meilleure compréhension de quelques paramètres opérant à de nombreuses échelles, mais aussi que la plupart des travaux se concentrent sur l'écoulement préférentiel dans la zone vadose et que très peu s'intéressent au sol de surface. L'écoulement préférentiel de l'eau peut avoir une incidence positive en contribuant à éloigner les crises, notamment au niveau de l'érosion due au ruissellement, de la genèse du sol en tant que service de l'écosystème, du recyclage des éléments nutritifs dans la zone racinaire et de la régulation générale de l'eau dans le cycle hydrique. Inversement, l'écoulement préférentiel peut avoir un impact négatif sur les problèmes éco-hydrologiques par le biais de la stabilité des pentes, du ravinement, des désastres géologiques, du traitement des eaux usées, de l'apport d'eau dans la zone racinaire et de la production d'aliments. Les auteurs concluent leur analyse en disant qu'on a besoin d'informations plus abondantes sur l'écoulement préférentiel avant de pouvoir en évaluer le rôle dans l'atténuation des crises de nature hydrique. [Traduit par la Rédaction]

Mots-clés : écoulement préférentiel de l'eau, eau, hydrologie du sol, sol.

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Introduction

Water is a key and integral part for the ecosystem services, and it is one of the complex components on the Planet operating at the interface of the hydrosphere, atmosphere, pedosphere, and biosphere. However, aridity, climate change, and extreme weather events pose severe threats to water sustainability (Maestre et al. 2015; Phalkey et al. 2015), which facilitate the concept of water crisis growing. An unprecedented water crisis, i.e., a concern over the amount and quality available to nations, has been predicted, and may already be underway (Elshafei et al. 2014). Heavily populated nations such as India and Pakistan share a common, but dwindling water resource (Fairless 2008; Tiwari et al. 2009; Hussain et al. 2018). Moreover, ecological degradation related with water crisis, such as desertification, groundwater contamination, pollutants movement, and flooding, is increasing in many regions of the world (van Oel and Hoekstra 2012; Cazcarro et al. 2014; Dalin et al. 2014; Elliott et al. 2014; Schewe et al. 2014; Lee and Bae 2015; Johansson et al. 2016; Michalak 2016; McColl et al. 2017). In these regions, existing water supplies are impacted by land-use practices. Imbalanced water flow pathways transport pesticides, fertilizer, and herbicides from the soil surface to the surface water body or the vadose zone, eventually polluting groundwater (Reichenberger et al. 2002; Kim et al. 2017; Villamizar and Brown 2017). The pathway for unwanted chemicals reaching groundwater is not well understood. In general, pesticides should be strongly detained onto soil particles. Even in systems with fine colloidal materials, it is expected that chemicals would still be better filtered by the surrounding soil matrix before reaching groundwater depths (Denver et al. 2009).

One possible pathway for this limited retention of chemicals is preferential water flow. Frameworks for evaluating the transport of pesticide and other contaminants via preferential water flow have been developed. Reichenberger et al. (2002) showed that preferential water flow was responsible for the major portion of pesticide transport in the soil profiles, and Kim et al. (2008) recorded that contaminant transport could be highly accelerated by preferential water flow despite that contaminants were detained by soil particles. Villamizar and Brown (2017) confirmed that preferential water flow was the main driver of pesticide transport at the catchment scale.

Preferential water flow refers to situations where in most (i.e., 70%–85%) of the water flow in the unsaturated zone travels via localized pathways and bypasses the surrounding soil matrix without limited resistance (Hendrickx and Flury 2001; Beven and Germann 2013; Zhang et al. 2016*a*). This process can influence large areas quickly, making it difficult to predict water flow in situ (Kapetas et al. 2014). Its origins are based on the discovery of soil macropores as continuous openings in the late 1800s (Schumacher 1864; Lawes et al. 1882; Engler 1919). Preferential water flow occurs not only in soils but also on the soil surface (Peñuela et al. 2016). It is now clear that preferential water flow in soils is more complex due to the high heterogeneity of the vadose zone. The vadose zone, where water, gas, and solute transfer between the soil surface and water table, is involved in many aspects of soil hydrology, such as soil water storage, soil erosion, water infiltration, evaporation, and groundwater recharge (Nielsen et al. 1986). Furthermore, preferential water flow on the soil surface is more related with soil erosion and other water-related environmental issues, such as debris flow, storm subsurface flow, and flooding (Peñuela et al. 2016). Peñuela et al. (2016) concluded that preferential water flow pathways on the soil surface occur as splash erosion and eventually make the soil surface smooth so that the spilling process is initiated in topographical depressions. Therefore, preferential water flow in soils and on the soil surface not only exerts significant effects on both surface water bodies and groundwater scarcity and security but also affects the soil physical hydrological processes to precipitation (Beven and Germann 2013; Khan et al. 2016). Moreover, quantification and characterization of preferential water flow is vital for groundwater recharge, waste disposal risk assessment, contaminants spatial pattern, and heavy metal redistribution in soils (Saravanathiiban et al. 2014). Preferential water flow could be better assessed and predicted in soil hydrology if there was wider consideration of the role of soil physics, soil chemistry, soil biology, and plant physiology.

However, our understanding of preferential water flow is incomplete. A better understanding of preferential water flow would benefit for understanding of the ecological and hydrological functions of soil, such as the transformation of substances and energy entering the soil, sorption of substances and water by soil, formation of chemical composition of groundwater, and the regulation of water balance of landscapes (Keesstra et al. 2014; Zhang et al. 2017). In this review, we begin with a focus on the scientific literature evaluating the improvement of preferential water flow then review the preferential water flow in terms of positive and negative roles in managing water. Finally, we identify some needed areas for research. The objective of this paper is to review the progress of preferential water flow with the aim of mitigating water crisis before they develop and to understand the contribution of preferential water flow to understand the landscape changes and the effect of management on the connectivity of the flows (Parsons et al. 2015; Masselink et al. 2017).

Preferential Water Flow Research Progress

Preferential water flow evaluation in comprehensive perspectives

Preferential water flow with high vertical infiltration rates originates from the variations in water flow

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velocity caused by high spatial heterogeneity of soil structure (Preti et al. 2018). In the previous studies even recent advances, a range of topics related with preferential water flow prompt our understanding for this research theme. For example, Helling and Gish (1991) distinguished root channels, cracks, and fissures as preferential pathways and concluded that those pathways could prompt water flow. Li and Ghodrati (1994) stated the influences of root channels on preferential transport of nitrates. Hagedorn and Bundt (2002) found that preferential pathways in forest soils could persist for decades. Jørgensen et al. (2002) reported that water flow patterns were more obvious in soil profiles with root channels than that without root channels. Šimůnek et al. (2003) summarized the models describing preferential water flow in vadose zone. Morris and Mooney (2004) stated that water flow recorded as instantaneous responses both in the 50 and 130 mm probes and showed that preferential water flow might be responsible for a significant portion of the total water flow (Mooney and Morris 2008). Franklin et al. (2007) showed that the mechanisms of preferential water flow were ambiguous. Allaire et al. (2009) reviewed the measuring techniques of preferential water flow in soils. Bogner et al. (2010) found that root biomass was larger in preferential pathways than that in the surrounding soil matrix. Gerke et al. (2010) reviewed the simulated models and measuring techniques of preferential water flow. Beven and Germann (2013) systematically reviewed preferential water flow and assessed the future trends and unresolved questions of preferential water flow. Etana et al. (2013) implied that persisted compaction of subsoil might enhance the strength of preferential water flow. Jarvis et al. (2016) reviewed the measuring methods of preferential water flow from pore to catchment scales. Xin et al. (2016) found that soils with distributed macropores prompted water flow increase remarkably. Zhang et al. (2016*a*) reviewed the development and progress of preferential water flow. Zhang et al. (2017) concluded that preferential water flow with high heterogeneity was more obvious in stony soils because of the complex interaction between plant roots and rock fragments.

Preferential pathways tend to be rich in the place where soil surface near-ponding conditions take place or where water flow encounters less permeable layers or where vegetation canopy architecture adjusts the redistribution of precipitation above ground surface (Bogner et al. 2013). Preferential water flow in cracked soils is a matter of very short time caused by crack closure, and this closure results from rapid and heterogeneous swelling processes (Liu et al. 2003; Wang et al. 2018). Water flow in cracks decreases when cracks are closed during wetting. However, some researchers stated that cracks did not close after rewetting, which could cause high water percolation during wetting. Cracks on the soil surface remain effective preferential pathways even after they are closed (Zhang et al. 2014). Stronger preferential water flow occurs in the fine texture soils or in the vicinity of trees than is found elsewhere or under ponding than under sprinkling irrigation conditions (Alaoui et al. 2011; Benegas et al. 2014), whereas weaker preferential water flow occurs in larger organic carbon content soils (Ghafoor et al. 2013). Zhang et al. (2017) stated that preferential water flow dye patterns were highly different in stony soils and that stony soils, especially the distribution of rock fragments, prompted the spatial heterogeneity of dye patterns.

Preferential water flow also occurs at following different land covers. (a) In agricultural soils, ploughing upper layers tend to be loose and uncompacted, but soil macropore connectivity and density decrease due to agricultural management (Koestel et al. 2013; Zhang et al. 2014; Jiang et al. 2017). Below the soil ploughing layer, the soil structure is compacted. Preferential water flow might occur because water flow encounters less permeable soil layers. Ploughing soil layers poses negative influences on soil fauna which reduces macropores number. Agricultural management practices, i.e., irrigation withdrawals, pose profound influences on the hydrological responses of the unsaturated vadose zone, in particular, the preferential transport of agricultural pesticides and pollutants (Perkins et al. 2011). Those pesticides and nutrients applied to agricultural soils could be conveyed by water flow to groundwater table. Preferential water flow pathways might be the driving force to transport those strongly adsorbed agricultural chemicals to soil depth (Allaire et al. 2011). Agricultural tillage practices also affect the transport of pesticides from the soil surface to groundwater table. Minimum tillage management practices, for example, keep the soil surface intact and improve the connectivity and number of preferential pathways (Keesstra et al. 2009; Di Prima et al. 2018). Those practices could reduce surface runoff generation and increase water percolation (Cerdà et al. 2018; Zhao et al. 2018). Connected preferential pathways control soil erosion rates (Cerdà et al. 2009; Cerdà and Doerr 2010; Rodrigo-Comino et al. 2018) such as the rock fragments do in agricultural soils (Rodrigo-Comino et al. 2017). Conversely, conventional tillage practices, i.e., plowing and harrowing, destroy most of the preferential pathways distributed in the topsoil layers, which increase surface runoff generation (Cullum 2009; Soto-Gómez et al. 2018). (b) In forest soils, a wide range of studies have been conducted to imply the mechanisms of preferential water flow (van der Heijden et al. 2013; Guo et al. 2014; Laine-Kaulio et al. 2014; Zhang et al. 2015*a*, 2015*b*; Julich et al. 2017; Zhang et al. 2017). Forest soils with lower bulk density in topsoil layers promote water vertical percolation because high interaction between preferential pathways and the surrounding soil matrix occurs (Alaoui et al. 2011). Forest soils are rich in large number of plant root systems and soil fauna (Noguchi et al. 1997; Höfer et al. 2001; Zhang et al. 2017). In contrast, root-induced channels in topsoil layers constitute preferential

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pathways in forest soils. Root-induced channels as roots shrink provide more space for water flow. Moreover, root-induced channels harbor high amounts of bacteria and are hot spots of bacteria activity (Dibbern et al. 2014). Larger and deeper plant root systems constitute a complex network within the soil profiles for perennial and long lived plants than annual and short-lived plants. Furthermore, decayed plant root systems could create more preferential channels to promote water flow (Yousefi et al. 2014). In contrast, soil fauna as a soil engineer and a driving force in nutrient cycling will improve soil carbon stocks and soil physicochemical properties (Höfer et al. 2001). Earthworms as preferential pathways are the most abundant in the primary forest soils. The primary forest soils include the highest soil fauna biomass, carbon stocks, and soil water content (Höfer et al. 2001). Soil fauna in forest soils could help maintain macropores connectivity (Noguchi et al. 1997). Meanwhile, macroporosity poses significant effects on soil faunal interactions with soil microbial activity (Vasconcellos et al. 2013). (c) In wetland soils, preferential water flow is different from agricultural and forest soils (Sade et al. 2010; Hua et al. 2014). From hydrological perspectives, wetland systems could connect shallow aquifers. During groundwater discharge, those systems, to some degree, could intercept contaminants and pollutants carried by water flow. The interception of wetland systems has been concluded both in laboratory experiments (Durst et al. 2013) and in field surveys (Narany et al. 2017). In general, promising interests in the study of preferential water flow and solute transport are focused on wastewater treatment through constructed wetland. During wastewater treatment, tracer experiments are used to check the type of preferential water flow, and the residence time of water flow in constructed wetland are applied gradually. In general, the residence time of contaminants and pollutants in preferential pathways is bigger than degradation time in constructed wetland (Wanko et al. 2009). (d) In grassland soils, preferential water flow could accelerate preferential solute transport to soil depth even groundwater table. Groundwater pollution is also more related with preferential water flow in grassland soils (Alaoui 2015). Although roots in grassland soils do not reach to deeper soil depth, groundwater contaminants still lead to serious environmental issues. Furthermore, water stored in deeper soils of grassland is not available for evapotranspiration (Alaoui 2015). For upper part of deeper soils in grassland, low to mixed (high to low) interaction between preferential pathways and the surrounding soil matrix dominates (Alaoui et al. 2011). However, in deeper soils of grassland, the network of preferential pathways mainly controlled bypass water flow (Weiler et al. 1998). Weiler and Naef (2003), by sprinkling and dye tracing experiments, concluded that continuous networks of macropores formed by earthworms in grassland soils dominated water flow process during extreme precipitation. Although it is the potential to be greatly worthy to investigate the effects of preferential water flow on groundwater contaminants in grassland soils, an indisputable fact is that low efficiency of preferential pathways of grassland soils in accelerating water flow downward to soil depth (Alaoui et al. 2011). Such could be explained as follows: (1) combined effects of finer and denser soil layers and the slope, (2) preferential pathways restricted numbers, (3) preferential pathways restricted connectivity and tortuosity, and (4) restricted interactions between preferential pathways and the surrounding soil matrix (Alaoui et al. 2011).

Preferential water flow evaluation in scale perspectives

The evaluation of preferential water flow in scale perspectives has been considered as an important research theme in recent years, implied by advances in useful methods (i.e., X-ray tomography) which improve imaging availability to better illustrate water flow process at multiple scales. Multiple scales, including the pore, field, catchment, and landscape scales, have become the research emphasis gradually (Jarvis et al. 2016). Advances of methods in measuring preferential water flow whether at smaller or larger scales have been stimulated.

Preferential water flow at smaller scales

Practical methods to study preferential water flow at the pore to core scales, pedon scales, and column scales are significantly focused in recent years. Preferential water flow in three dimensions is increasingly quantified by X-ray tomography. This technique could capture the topological and geometrical properties of the soil pore space network (Jarvis et al. 2016). And such could also describe the spatial properties of water and solute in the soil pore space, especially in relation to preferential solute transport at the pore to core scales (Koestel and Larsbo 2014). Apart from X-ray tomography, magnetic resonance imaging, positron emission tomography, neutron radiography, and single photon emission computed tomography are also applied to the study of preferential water flow and solute transport (Jarvis et al. 2016). Badorreck et al. (2012), by means of X-ray tomography and neutron radiography, stated that preferential water flow driven by gravity through burrow occurs in initially dry soil conditions, whereas the burrow could not convey any water flow in initially wet soil conditions.

Apart from those promising techniques above, breakthrough curves are also widely used to reveal preferential water flow in laboratory analyses. Recent years, researchers tend to evaluate the controlling factors of preferential water flow by various breakthrough curves shape (Jarvis et al. 2016). In general, breakthrough curves shape could reflect the strength of preferential water flow; however, it could not imply the spatial pattern of water flow in soil columns. Köhne et al. (2006) stated that

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preferential water flow at column scales could not be predicted efficiently on the basis of water flow and solute concentration parameters. Luo et al. (2010) concluded that the strength of preferential water flow was weaker in soil columns with larger macroporosity. Larsbo et al. (2014) stated that smaller macroporosity in soil columns had smaller near saturated hydraulic conductivity which resulted in the occurrence of preferential water flow.

Moreover, for field soil profile scales, the most convincing evidence of preferential water flow is characterized by dye tracing experiments (Flury et al. 1994; Hagedorn and Bundt 2002; Köhne et al. 2006; Backnäs et al. 2012; Bargués Tobella et al. 2014; Mossadeghi-Björklund et al. 2016; Zhang et al. 2017). Hagedorn and Bundt (2002) and Backnäs et al. (2012) used dye tracing experiments to compare the difference of soil chemical properties between preferential water flow pathways and the surrounding soil matrix. Gimbel et al. (2016) also used this method to imply the effects of climate changes on the strength of preferential water flow in forest soils. However, dye staining patterns may not reveal all preferential water flow pathways of soil profiles (Beven and Germann 2013), and it only shows water flow patterns in two dimensions. In particular, dye tracing experiments are time consuming.

Preferential water flow at larger scales

Preferential water flow at larger scales (i.e., field plot, hillslope, catchment, and regional scales) could be reflected by space–time variations in soil moisture or solute concentration (Jarvis et al. 2016). The occurrence of preferential water flow at the field plot scales is illustrated. For example, Wessolek et al. (2009) stated the mechanism of changes of preferential water flow patterns with seasonal soil water repellency at the plot scales. Yang et al. (2013) explored various controlling factors of preferential water flow within a 144 m^2 field plot scales. Hillslopes act as filters for water flow to the surface water bodies or even to the groundwater table. Preferential water flow alongside hillslopes is highly related to morphological properties like low permeability horizons in soil and compacted bedrock (Beven and Germann 2013). Meanwhile, individual preferential water flow pathways could extend across large distance along the hillslope (Jarvis et al. 2016). A wide range of studies conclude that preferential water flow might be crucial in controlling hillslope hydrological processes. Interests in the role of preferential water flow in hillslope hydrology have increased dramatically (Leaney et al. 1993; Guebert and Gardner 2001; van Schaik et al. 2008; Anderson et al. 2009; Klaus et al. 2013; Guo et al. 2014). Noguchi et al. (1999), for example, implied that preferential water flow pathways formed by living or decayed roots and bedrock fractures might be important in determining hydrological processes at hillslope scales. Anderson et al. (2009) stated that preferential water flow pathways carried most water flow during storms at hillslope scales. If preferential water flow is crucial at hillslope scales then it is also for catchment scales (Beven and Germann 2013). Wessolek et al. (2009) investigated the variations of preferential water flow seasonally at larger scales. Liu and Lin (2015) found that preferential water flow was widely controlled by soil type and topography on the basis of soil moisture data at catchment scales. A promising technique to assess space–time dynamics of preferential water flow at catchment scales is the use of sensor response times (Hardie et al. 2013; Liu and Lin 2015; Wiekenkamp et al. 2016). Christiansen et al. (2004) found that preferential water flow affected more contaminants leaching into the groundwater table at catchment scales though preferential water flow had no dominating influences on groundwater recharge or discharge. Wiekenkamp et al. (2016), by using sensor response times, concluded that preferential water flow was triggered by the amount and intensity of precipitation at catchment scales. However, preferential water flow at catchment scales requires more investigation because of the high temporal and spatial heterogeneity of water flow dynamics (Allaire et al. 2009; Beven 2010; Darracq et al. 2010; Godsey et al. 2010; Beven and Germann 2013; Wiekenkamp et al. 2016). The geometry, connectivity, and tortuosity of preferential pathway networks remain an issue at catchment scales (Ghafoor et al. 2013). Direct tracing of preferential water flow remains a question even at zero-order catchment scales. Moreover, public authorities need integrated larger-scale models and decision-support tools to decide the effects of preferential water flow at catchment scales. In final, at regional scales, spatial variations in climate could affect the occurrence of preferential water flow. McGrath et al. (2009), by preferential water flow index method, studied the influences of soil properties and rainfall on solutes leaching within regional scales. Steffens et al. (2015), by preferential water flow modeling methods, evaluated the effects of climate changes on solutes leaching to the groundwater table at regional scales.

Preferential Water Flow Interaction with the Surroundings

The interaction between preferential water flow pathways and the surrounding soil matrix affects the ability of water transfer and solute transport (Laine-Kaulio et al. 2014; Williams et al. 2016). At the primary stage, water transfer mainly occurs from preferential pathways into the surrounding soil matrix because of the higher water content in preferential pathways. Peterson and Wicks (2005) concluded that only 1% of the volume of water and <<1% of the amount of solute moved from preferential water flow pathways to the surrounding soil matrix. As time goes on, extent of interaction between the two zones increase (Bogner et al. 2012), and water transfer is typically common. At the same time, the surrounding soil matrix exchanges the old infiltrating

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water with new water during successive water infiltration (Mohanty et al. 2016). And preferential water flow could also displace those older water stored in the surrounding soil matrix deeper into the soils. In the final stage of infiltration, the phenomenon of hysteresis occurs when the surrounding soil matrix functions significantly. Mohanty et al. (2016) concluded that extent of hysteresis was stimulated with increases in exchange of water and solute between preferential water flow pathways and the surrounding soil matrix. However, in wet soil conditions, preferential water flow is enhanced due to the reduced lateral water flow from the preferential water flow pathways to the surrounding soil matrix (Hardie et al. 2013; Alaoui 2015).

The interaction between preferential water flow pathways and the surrounding soil matrix is greatly influenced by soil properties and soil water content (Weiler and Naef 2003). A high interaction between the two zones which results in more water storage in soils is caused by the permeability of surrounding soil matrix and low initial soil water content. Moreover, a low interaction which causes water bypassing the surrounding soil matrix is generated due to the low permeability of the matrix or the nearly saturated matrix (Weiler and Naef 2003). Both the interaction between the two zones and the permeability of underlying bedrocks dominate water flow in preferential pathways. If the interaction is low and the permeability is high, the water flow in preferential pathways is high. In contrast, if both the interaction and the permeability are low, water flow in preferential pathways is reduced and the saturation of preferential pathways occurs (Weiler and Naef 2003). The interaction between preferential pathways and the surrounding soil matrix is concluded in Table 1.

Moreover, interaction between the two zones is also related with soil texture, soil organic matter content, and land coverage type (Jarvis 2007; Alaoui et al. 2011; Backnäs et al. 2012; Zhang et al. 2017). For example, soil properties, i.e., soil organic matters, may be different between the two zones (Bogner et al. 2012). The higher soil organic matter content in preferential pathways than the surrounding soil matrix is attributed to the main sources as follows: (1) Greater proportion of living or decayed roots in preferential pathways than the surrounding soil matrix. (2) Preferential input of dissolved organic matter from the soil surface. (3) Enhanced release of microbial biomass carbon during rewetting process of dry soil (Morales et al. 2010). (4) Activity of soil fauna dominated by earthworms, termites, and ants increasing the quality of soil carbon stock, driving the cycling of nutrient, and high nutrient release of earthworms activity (Höfer et al. 2001; Bertrand et al. 2015).

Preferential Water Flow Inconsistent Role in Ecosystem Services

Natural resources that support the sustainable development and health of ecosystem services have been reviewed as one of the most important elements of capital. It is clearly evident that water as the earth's natural capital could transport through soil profiles preferentially. In the sense of soil's value, the transport of gas, energy, materials, water, and other solutes could be influenced by soil natural characteristics directly (Clothier et al. 2008). Such is viewed as soil's ecosystem services. The soil's ecosystem services, such as preferential water flow of solutes (energy and materials), might have inconsistent role due to the occurrence of preferential water flow. However, it is evident that studies have conducted limited assessments to state the inconsistent (positive/negative) role of preferential water flow in environments.

An emerging topic of preferential water flow in recent years has been focused on the negative effects of preferential water flow on environments, whereas the positive role of preferential water flow tends to be neglected. Further investigations of preferential water flow's inconsistent influences on ecosystem services should be stimulated to better understand why most concerning environmental, economic, and social issues are relevant to preferential water flow (Clothier et al. 2008). In this paper, we have considered 10 ecosystem services commented by Costanza et al. (1997), and we also have discussed the inconsistent role of preferential water flow played across those ecosystem services. We suppose that these discussions will make a sense for future studies of how preferential water flow could enhance/diminish ecosystem services (Clothier et al. 2008).

Preferential water flow positive role

Runoff generation control

Preferential water flow pathway openings might constitute a finite portion of soil layers, but those pathways at the land surface could convey more water to depth, which results in more percolation and low interaction with the surrounding soil matrix. Alaoui et al. (2011) used dye tracer experiments to conclude that forest soils might be more efficient to prevent runoff generation at the land surface than grassland soils. Such is caused by more connected preferential pathways formed by plant roots in forest soils. Those preferential root channels in topsoil layers prompt more water percolation to depth. Meanwhile, soil water repellency is also a crucial element which is more related with runoff generation. It is clearly known that few runoff generation occurs when soil water repellency is lower. For example, Alanís et al. (2017) found that runoff generation could be prevented as soil water repellency is usually lower, which leads to the increase of soil water infiltration capacity and the promotion of preferential water flow pathways. However, deeper preferential pathways will not be necessarily generated if higher soil water repellency occurs (Rye and Smettem 2017). Rye and Smettem (2017) noted that though average preferential pathways depth increased with soil water repellency, more shallower

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Table 1. The interaction between preferential pathways and the surrounding soil matrix indicated by preferential pathways width and dye coverage (Weiler and Flühler 2004).

Interaction between preferential pathways and surrounding soil matrix	Preferential pathways width <20 mm	Preferential pathways width >200 mm
	Dye coverage (%)	Dye coverage (%)
Preferential water flow with low interaction with surrounding soil matrix	>50	<20
Preferential water flow with mixed interaction (high and low) with surrounding soil matrix	20–50	<20
Preferential water flow with high interaction with surrounding soil matrix	<20	<30
Heterogeneous soil matrix flow and fingering	<20	30–60
Homogeneous soil matrix flow	<20	>60

preferential pathways were generated in high soil water repellency than that in low or medium repellency conditions. And they also found that preferential pathways were more wider in very strongly repellent soils.

Soil formation development

Soil physicochemical and biological properties could be highly improved by preferential pathways, and those properties might be different between preferential water flow pathways and the surrounding soil matrix (Bogner et al. 2012). In general, preferential pathways as a sink or storage of organic matters, microorganisms, and bacteria (Garrido et al. 2014; Zhang et al. 2016*b*) pose significant effects on the soil formation. For example, soil bulk density, structure, macroporosity, and permeability could benefit from the positive feedback of preferential pathways (Cerdà 1996; Bochet 2015; Singh et al. 2015). Well-connected preferential pathways in soils increase macroporosity connectivity and density which prompt soil permeability. This in turn benefits for soil structure development. Those preferential pathways prompt the transfer of water, solute, energy, and gas to soil depth. Moreover, drivers of soil formation, i.e., land use, soil fauna, vegetation, climate, and topography, could in turn influence the strength and occurrence of preferential water flow (Jarvis et al. 2012).

Nutrient cycling in root zone

The service of nutrient cycling in root zone benefits from preferential water flow (Clothier et al. 2008). We discuss, as an example, that inorganic nitrogen could be rapidly leached from the root zone through preferential water flow pathways when nitrogen mineralizes in the surrounding soil matrix. However, mineralized nitrogen is still accessible to plant roots (Clothier et al. 2008). The root zone is rich in root channels and soil microorganisms. Those channels including abundant organic matters could in turn prompt plant root growth. Moreover, preferential root channels longevity might be improved by the increase of soil aggregate stability which is supported by the increase of soil organic matter content around root systems in root zone (Ghestem et al. 2011). Meanwhile, plant roots themselves also decompose and release some organic matters into the soil (Ghestem et al. 2011). In particular, at the onset of senescence, preferential root channels are filled with abundant organic matters released by root itself (Ghestem et al. 2011).

Plant root systems not only grow into macropores but also constitute lots of preferential root channels themselves. Macropores have higher amounts of nutrients and organic matters, and are regarded as storage of microorganisms compared with the surrounding soil matrix. The nutrient cycling in root zone is more efficient because preferential water flow pathways could be persistent for decades (Hagedorn and Bundt 2002).

Water regulation

In physical hydrology, the infiltrating surface water and solute could quickly get into the groundwater table through the small fraction of preferential water flow pathways (Hendrickx and Flury 2001). Upward flow from groundwater into the surface water is also present (Kalbus et al. 2006; Schmidt et al. 2006). Groundwater and surface water interaction poses a significant influence on water regulation and balance (Pahar and Dhar 2014). Preferential water flow could affect hydrological responses to precipitation and, therefore, both groundwater and surface water management (Jarvis 2007; Clothier et al. 2008; Beven and Germann 2013; Jarvis et al. 2016). According to surface water, preferential water flow pathways could occur on the soil surface near ponding conditions, whereas for groundwater zone, those preferential pathways could occur when water flow encounters less permeable soil layers (Jarvis et al. 2016).

Preferential water flow negative role

Slope stability control

It is well known that subsurface drainage systems are necessary to reduce subsurface pores water pressure.

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Subsurface pore water pressure is more related with slope stability. We could not deny the fact that pore water pressure might increase and concentrate in critical zone alongside the slope as subsurface water flow pathways terminate in the slope (Ghestem et al. 2011). It is noted that preferential water flow is crucial part of subsurface water flow systems. Preferential water flow pathways are dramatically treated as conduits of subsurface water flow in the slope. Preferential pathways could feed water to make pores water pressure increase. When the landslide develops, the water storage continues to feed water even drainage occurs. Subsurface water flow may develop to accelerate as landslide happens (Hencher 2010). Such will prompt slope instability.

Subsurface water flow is a combination of preferential water flow and soil matrix flow. Leaney et al. (1993) concluded that >90% subsurface water flow was mainly carried by preferential water flow pathways. Anderson et al. (2009) stated that subsurface water flow velocity was governed by the connectivity of preferential water flow pathways. Stumpp and Maloszewski (2010) found the contribution of preferential water flow pathways to subsurface water flow was between 1.1% and 4.3% for the lumped parameter approach and 1.1% and 20.5% for the HYDRUS 1-D approach, respectively. Vogel et al. (2010) found that only 24% subsurface water flow was transported through preferential water flow pathways. Rapid subsurface water flow depends on the fluxes of both vertical and lateral preferential water flow pathways (van Schaik et al. 2008; Nieber and Sidle 2010). Meanwhile, repaid subsurface water flow through preferential water flow pathways could affect hillslope stability and landslides. These negative impacts of preferential water flow on ecosystem services are clearly discussed. Preferential water flow located in subsurface water flow systems might result in the accumulation of sediments in pipes, which leads to blockage, underground erosion, and damage to surface infrastructure caused by sinkholes (Øygarden et al. 1997).

Gully erosion control

Preferential water flow through soil-pipe gullies poses a negative role in the gully erosion control. Due to the higher shear forces compared with frictional strength, repaid preferential water flow is more easier to erode the periphery of the soil pipes (Wilson et al. 2008). Binding soil particles becomes difficult when pipe erosion occurs. Considering pipe erosion developing, gully erosion is prompted when soil pipes collapse (Faulkner 2006). Preferential water flow through soil pipes results in internal erosion of soil pipes, which may prompt the development of gullies (Wilson et al. 2008). In European areas, almost 60% of gully erosion events are caused by preferential water flow through soil pipes (Bocco 1991).

Geological disaster control

Geological disaster (i.e., debris flow and floods) might be promoted by preferential water flow pathways under high precipitation conditions. The increase of geological disaster is attributed to the faster transfer of rainfall to preferential water flow pathways located aboveground soil surface. Heavy rainfall in excess of the soil surface water-holding capacity will help promote the geological disaster events. Taking debris flow as an example, it is clearly evident that sheet or splash erosion could make the soil surface smoothly by eroding clods to flat the roughness and eventually to fill the depression with the sediments, which leads the increase of surface runoff and the decrease of storing water in the depression during high precipitation. In particular, once the spilling process is initiated in the depression, preferential water flow pathways occur among these depressions alongside the steep slope. Preferential water flow pathways on the soil surface could prompt, route, and concentrate the surface runoff, sediment, gravel, and stone transport with high velocity during high precipitation (Peñuela et al. 2016). Thus, debris flow occurs quickly with the synthesis of turbulent surface runoff sediment and gravel interacts.

The second common event, floods, could also make serious disaster to people's life and health. In general, floods with high heterogeneity at catchment scales could displace those surface water bodies (i.e., rivers, lakes, and reservoirs) to the downstream. The connectivity of surface water bodies within the catchment may prompt the understanding of preferential water flow negative role in the flooding events. Water flow connects different surface water bodies of different landscapes. Water flow prompts the exchange of water, solute, and energy between these water bodies. As floods occur during high precipitation within the catchment, those water bodies with high hydrological connectivity could interact to make the flooding event more serious. The hydrological connectivity between different water bodies could be treated as an index of preferential water flow pathways.

Waste treatment

It is clearly noted that wastes conveyed to the water bodies will make serious pollution. Those wastes, i.e., domestic and industrial wastewater, garbage, food residue, and contaminants, might be difficult to be removed or degraded from water bodies if they have been conveyed by water flow. In particular, preferential water flow carries those wastes to the groundwater table quickly. For example, it is imperative for hydrologists to understand why pesticides and contaminants could be reported at depths greater than would be expected. Groundwater table polluted by unwanted contaminants will result in extensive impacts on drinking water systems and local economies. A wide range of studies have been conducted to quantify the effects of preferential water flow on contaminants transport by models

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(Reichenberger et al. 2002; Kim et al. 2008; Villamizar and Brown 2017). Those studies could ahead clarify the origin of groundwater pollution.

Water supply in root zone

A negative role of preferential water flow in the service of water supply in root zone is depending on the leakiness of water flow through those connected preferential pathways (Clothier et al. 2008). Plant roots preferentially absorb water from macropores then from micropores of the surrounding soil matrix. In the dry season, soil cracks appear because of swelling and shrinkage conditions, conversely, in the cold season, cracks appear caused by freeze–thaw processes (Ghestem et al. 2011). Those cracks convey water to depth preferentially. However, plant root systems might not absorb water conveyed by cracks to support its sustainability. Though some cracks might be over-capillary, water flow still occurs along the walls of those cracks after soil ponding (Gerke 2006). Apart from the cracks caused by climate changes, preferential root channels could also prompt water to depth. Root channels development in root zone, in particular for longer and wider channels, could route water away from unstable zones and convey water in the slope efficiently (Ghestem et al. 2011). In general, forks of root channels might divide even concentrate water flow. It is clearly noted that more preferential root channels alongside the slope, more water drainage is stimulated in this direction. Compared with live plant roots, those dead root channels could increase inter-connectivity among root systems (Ghestem et al. 2011). Preferential root channels might dissipate pore water pressure to promote water flow away from the root zone because these root channels mainly occur in the shallow soil layers (Ghestem et al. 2011). Meanwhile, plant roots could grow through the soil profiles even the bedrock and new openings of void spaces generated by root growth (Liu et al. 2018), which all lead to the lose of water systems. In total, preferential water flow is negatively correlated with water supply in root zone which leading low water conservation and availability during plant growth.

Food production

Plant roots growth need more water and nutrients to sustain their development. The root zone rich in organic matters and soil moisture content favors in the development. Those well-connected preferential pathways in topsoil tend to prompt available water to depth. A question is that plant root systems not only provide preferential root channels to stimulate water flow along the wall of channels but also conserve soil and water. How to distinguish the two aspects functions, it is still needed. However, it is a fact that well-connected preferential pathways pose negative effects on water supply in root zone, as noted above, which could prevent crop growth and decrease food production. In particular, according to agricultural soils in dry seasons, large cracks are gradually generating on the soil surface during shrink–swelling cycles. In cold season, cracks appeared by freeze–thaw processes (Ghestem et al. 2011). During crop growth, those cracks as preferential pathways or macropores make water loss directly. Crop roots could not absorb more water to keep their balance, which is adverse for food production. Preferential water flow might result in reduced agricultural production (Cullum 2009; Garg et al. 2009).

Preferential Water Flow Future Challenges

The study of preferential water flow in soil science has emerged as a significant research theme in recent years. The contributions of this special review include the progress of preferential water flow in the vadose zone and the advanced development of discussion on preferential water flow on the soil surface. However, despite the growing interests of preferential water flow worldwide, a real opportunity and challenge of preferential water flow studies in future occurs for hydrologists. Given the unpredictable significance of preferential water flow related to groundwater resources security and agricultural nonpoint source pollution, a wide range of unresolved questions and disputes still confuse researchers (Beven and Germann 2013).

I make the following suggestions as areas of research:

- A new theory of preferential water flow, especially incorporating hydrological models efficiently, needs to be rigorously tested. An integration of all the purported drivers including preferential water flow into hydrological models is required. A fully convincing integrated physical theory into models still has not been achieved (Beven 2010). Though models describing water flow process through preferential water flow pathways have advanced rapidly during the past decades, such as dual-permeability model, dual-porosity model, dual-continuum model, and HYDRUS-2D model, models for preferential water flow are still limited at the hillslope even catchment scales (Gerke et al. 2010). Meanwhile, current confusions of modeling preferential water flow still exist incorporating smaller scale simulations into the larger scale. It is noteworthy that preferential water flow is generally ignored in lumped hydrological models, even in physically based models (Weiler 2017). An active field for future research needs to extend, develop, and assess the effects of preferential water flow on such models sensitivity at multiple scales. In recent advances, how to make preferential water flow model parameterization? It is lacking.
- A review of preferential water flow studies in the last decades indicates that we only confirm the effects of existing preferential water flow pathways in the vadose zone. Existing preferential water flow

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pathways constructed by X-ray computed tomography could be easily quantified by means of macropores characteristics, including density, number, connectivity, tortuosity, length, width of macropores. However, those potential preferential water flow pathways could also be activated to accelerate water flow velocity. Splash erosion associated with high precipitation, for example, erodes sediments from the soil surface over time, and afterwards a depression alongside the hillslope occurs. The depression in turn causes water infiltration into soils. How to quantify and characterize the effects of those potential preferential water flow pathways on hydrological processes? It is more complex and greatly needed.

- Other questions remain: How preferential water flow pathways carry water flow in the vadoze zone? How water in preferential water flow pathways interact with water in the surrounding soil matrix? How solute exchange between preferential water flow pathways and the surrounding soil matrix? How important preferential water flow pathways are in terms of water flow at the hillslope or catchment scales? How solute process adsorption and degradation on preferential water flow pathways walls?
- Whether preferential water flow matter at the catchment scales or not? Preferential water flow phenomenon occurs under specific conditions linked with soil water content, soil properties, and high precipitation. However, these conditions could not assess hydrographs in the long run due to the small fraction of preferential water flow time. Hydrologists typically characterize preferential water flow at the local scale, and more consideration is needed at the scale of the catchment.
- Economic valuation of the effects of preferential water flow on ecosystem services remains a challenge, and it will increase with storm periodicity and intensity. Clothier et al. (2008) calculated that preferential water flow in terrestrial ecosystems is globally worth US$304 billion per year. How many economic valuations could be better assessed for the positive role of preferential water flow on environments? And how many for the negative role? In summary, preferential water flow studies in the vadose zone should not be the only research theme for hydrologists; however, preferential water flow studies on the soil surface should also be fully considered when people assess the effects of preferential water flow on ecosystem services.
- Hydrological connectivity of preferential water flow pathways at the land surface or in the vadose zone is a crucial research theme in recent years. Understanding the role of hydrological connectivity and integrating the concept of hydrological connectivity into hydrological models could benefit for the policy makers to know how water flow connects between or among different water bodies. Hydrological connectivity positively benefits for the exchange of mass and energy between different water bodies. However, what remains to be settled is the properties and size of the connected preferential pathway networks.

Conclusions

A large number of studies reviewed in this paper show that there are promising developments in our understanding of preferential water flow, but it is evident that some important issues remain unresolved. Most studies focus on preferential water flow in the vadose zone, whereas studies of preferential water flow on the soil surface are greatly lacking. It is evident from the literature that the positive benefits and negative impacts of preferential water flow on ecosystem services need to be better understood. This review concludes that the research field of preferential water flow both in the vadose zone and on the soil surface needs to work towards an integrated framework to evaluate its inconsistent role in mitigating water-related crisis events.

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